

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element C Initial Template - AT - 18 points

The requirements from Element C will be used for Elements D, E, F, and H.

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

We will make the hydrogen car more accessible to more of the country because we will make multiple chargers around America, and we will research making the hydrogen-powered car, and also add modifications that can keep the hydrogen gas from leaking, making the chargers not cost much to produce

It ties to the prior research from Element A because it uses sources on the negative consequences of gasoline-powered cars. In Element B, my team is improving the hydrogen-powered car design by collecting more hydrogen in the air to make the car run longer, and also using a turbine that uses the air rushing in to go farther. We will research these topics to see if these are capable achievable experiments

The design requirement was __unapproved_____ (approved/ unapproved) by the stakeholder. Because he was unavailable at that time.

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Design Requirement #2 (HIGH Priority)

A system that takes the hydrogen from water and stores it safely, and doesn't blow up in our face. Or malfunctions and leaks hydrogen and could combust in your car or harm and family members. To find a solution to this problem is to make a tank that can hold the hydrogen without leaking. Hydrogen stations are usually located at existing gas stations and dispense hydrogen as a compressed gas, similar to compressed natural gas. Hydrogen dispensers are similar in appearance to retail fuel dispensers. Hydrogen is delivered at 35 MPa or 70 MPa (5,000 psi or 10,000 psi) to a vehicle's tank. <https://www.ourair.org/wp-content/uploads/THRPAppendixC6.pdf>

This requirement directly ties the prior research to element A or B because we are both trying to make it more accessible to the people of the north, south, east, and west

The design requirement was ____approved____by my dad_____ (approved/unapproved) by the stakeholder.

Design Requirement #3 (HIGH Priority)

The design requirement for our design is to make the charging stations more accessible to the hydrogen-powered cars. Also, to help make the cars not travel so far just to find a charging platform.



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The design requirement directly ties to our prior research on A, on how we should make the hydrogen-powered stations more accessible to the public. The solution to this problem would to make sure these charging stations aren't super expensive to produce and to use

The design requirement was __ __ approved _____ (approved/ unapproved) by the stakeholder.

Design Requirement #4 (HIGH Priority)

A way to make hydrogen continuously using water with energy that is safe, constant, and safe for the environment. Solar only works with direct sunlight, wind only works when there is wind, and nuclear has radioactivity.

In order to make hydrogen cars we need fuel that is able to be used safely. The fuel must be in a liquid form through any means (pressure, cyrogenics, etc)



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The design requirement was approved (approved/ unapproved) by the stakeholder.

Design Requirement #5 (HIGH Priority)

We need to create a proper cryogenic storage unit to supercool the hydrogen into liquid hydrogen for the fuel cell system. The system must be efficient and safe using less power than the system creates.

It is important because the design shouldn't explode violently, and also because if it uses too much power (more than can be created) it wouldn't even get on to the highway.

The design requirement was approved (approved/ unapproved) by the stakeholder.
The stakeholder said th

Design Requirement #6 (LOW Priority)

We qq will create a magnetic-powered engine that utilizes magnetic force and a gear system to move the magnet back and forth, rather than keeping it stationary, as this will cause the magnetic force to even out and come to a stop, much like glue.

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This ties back to the prior research from Element B because it shows a previous design of a magnetic engine. While it used electricity, it required a lot of energy, necessitating frequent charging.

The primary stakeholder is ML's dad, who works with energy

Design Requirement #7 (LOW Priority)

The design requirement was _____ (approved/ unapproved) by the stakeholder.

Rubric

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Element C: Presentation and Justification of Solution Design Requirements

C1. Determination of design requirements. _____ design requirements are related to prior research.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	No
C2. Requirements list. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	are listed, prioritized, clear, and detailed	are listed, prioritized, generally clear, and detailed	are listed, prioritized, generally clear, and somewhat detailed	are listed and prioritized, but some or all may be incomplete and/or lack specificity	are listed, formatted, and prioritization is attempted, but these may be partial or overly general	are not present or are too vague
C3. Quantity of high-priority design requirements. _____ of the design requirements listed are identified as high priority.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	Five	Four	Three	Two	One	None
C4. Objective & measurable. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	are consistently objective and measurable	are nearly always objective and measurable	are generally objective and measurable	may be only sometimes objective and/or measurable	are insufficiently measurable	are not measurable
C5. Leading to solution. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	would be highly likely to lead to a tangible and	would be likely to lead to a tangible and viable	has the potential to lead to a tangible and	has some potential to lead to a tangible and	have little potential to lead to a tangible and	cannot lead to a tangible and viable problem



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	viable problem solution	problem solution	viable problem solution	viable problem solution	viable problem solution	solution
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	no primary stakeholder groups.